

STM32 node architecture

Local services

mini9P | RPC | Job | /sys/time | /compute

Service runtime + compute worker

protocol dispatch on service task; bounded work on low-priority compute task

Node control

address | lease | routes | relay | wall offset

Port manager

HELLO FSM | peer identity | health

UART DMA transport

IDLE RX | frame pool | parser | priority TX queues

Fixed tasks, fixed frame blocks and no dynamic allocation in the communication path.